

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Bleed Air Heating and Ventilation System	105-80094	22.2	2300	51060
Bleed Air Heating and Ventilation System	105-80097	22.2	2300	51060
Bleed Air Heating and Ventilation System	105S-80095	(22.3)	(2200)	(49060)
NOTE Above values in parentheses apply to Model S Variants.				

7. SYSTEM DESCRIPTION

The heating and ventilation system (see Fig. 1) comprises a bleed air heating subsystem and a fresh air ventilation subsystem having a common air distribution ducting. The subsystems may be operated separately or together.

Air is distributed via the common ducting to outlets throughout the cabin; two windshield manifolds with individual nozzles for the cockpit area and two individual nozzles each for the passenger and cargo areas. The individual air nozzles are adjustable in both flow rate and direction.

7.1 VENTILATION

The cabin may be ventilated with fresh air provided from a ram air inlet on the lower fuselage nose shell. A notched lever assembly on the lower right of the instrument panel operates a valve in the distribution ducting to control the amount of fresh air entering the cabin and, when placed in the BLWR position, additionally activates a blower fan.

7.2 BLEED AIR HEATING

Hot high-pressure bleed air, tapped from each engine's diffuser scroll, is mixed with outside air to obtain a desired temperature and then distributed via the common ducting to the cabin outlets.

Normally the temperature control operates automatically to regulate the selected temperature by sensing the air temperatures within the ducting and cabin area, and then adjusting the temperature control valve accordingly. Should the automatic control malfunction adjustment of the valve may then be performed manually by means of controls provided.

Bleed air heating controls are on the overhead console switch panel. A three-position heating control toggle switch labeled MAN-AUTO and having a center off position is used